

Exp No: 5

Arduino based manual electronic counter using proteus

AIM:

To write an Embedded C program for Manual Electronic counter using Arduino IDE and Proteus

SOFTWARE REQUIRED:

- Proteus 8 software
- Arduino IDE

PROGRAM:

```
int x0,x1,x2,x3,x4,x5,x6,x7,x8,x9;
```

```
int delay_time=200;
```

```
void setup() {
```

```
//configure pin2 as an input and enable the internal pull-up resistor
```

```
pinMode(12, INPUT_PULLUP);
```

```
pinMode(1,OUTPUT);
```

```
pinMode(2,OUTPUT);
```

```
pinMode(3,OUTPUT);
```

```
pinMode(4,OUTPUT);
```

```
pinMode(5,OUTPUT);
```

```
pinMode(6,OUTPUT);
```

```
pinMode(7,OUTPUT);
```

```
}
```

```
void loop() {
```

```
//read the pushbutton value into a variable
```

```
int sensorVal = digitalRead(12);
```

```
if (sensorVal == LOW) {
```

```
x0=true;

}

while(x0){
    zero();
    sensorVal = digitalRead(12); if
    (sensorVal == LOW) {
        x1=true;
        x0=false;
    }
}

while(x1){
    one();
    sensorVal = digitalRead(12); if
    (sensorVal == LOW) {
        x2=true;
        x1=false;
    }
}

while(x2){
    two();
    sensorVal = digitalRead(12); if
    (sensorVal == LOW) { x3=true;
    x2=false;
    }
}

while(x3){
    three();
    sensorVal = digitalRead(12); if
    (sensorVal == LOW) { x4=true;
```

```
x3=false;

}

}

while(x4){
four();
sensorVal = digitalRead(12); if
(sensorVal == LOW) { x5=true;
x4=false;
}
}

while(x5){
five();
sensorVal = digitalRead(12); if
(sensorVal == LOW) { x6=true;
x5=false;
}
}

while(x6){
six();
sensorVal = digitalRead(12); if
(sensorVal == LOW) { x7=true;
x6=false;
}
}

while(x7){
seven();
sensorVal = digitalRead(12); if
(sensorVal == LOW) { x8=true;
```

```

x7=false;

}

}

while(x8){

eight();
sensorVal = digitalRead(12); if
(sensorVal == LOW) { x9=true;
x8=false;

}

}

while(x9){
nine();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x0=true;
x9=false;

}

}

}

void zero()

{
for(int i=1;i<7;i++)
{
digitalWrite(i,HIGH);
digitalWrite(7,LOW);
}
}

```

```
delay(delay_time);  
}
```

```
void one()  
{  
digitalWrite(1,LOW);  
digitalWrite(2,HIGH);
```

```
digitalWrite(3,HIGH);  
digitalWrite(4,LOW);  
digitalWrite(5,LOW);  
digitalWrite(6,LOW);  
digitalWrite(7,LOW);  
delay(delay_time);  
}
```

```
void two()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,LOW);  
digitalWrite(4,HIGH);  
digitalWrite(5,HIGH);  
digitalWrite(6,LOW);  
digitalWrite(7,HIGH);  
delay(delay_time); }
```

```
void three()  
{  
digitalWrite(1,HIGH);
```

```
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,LOW);  
digitalWrite(6,LOW);  
digitalWrite(7,HIGH);  
delay(delay_time); }  
  
void four()  
{  
digitalWrite(1,LOW);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,LOW);  
digitalWrite(5,LOW);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time);  
}  
  
void five()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,LOW);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,LOW);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time); }  
  
void six()
```

```
{  
digitalWrite(1,HIGH);  
digitalWrite(2,LOW);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,HIGH);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time);  
}
```

```
void seven()
```

```
{  
digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,LOW);  
digitalWrite(5,LOW);  
digitalWrite(6,LOW);  
digitalWrite(7,LOW);  
delay(delay_time); }
```

```
void eight()
```

```
{  
digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,HIGH);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);
```

```

delay(delay_time);
}

void nine()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,LOW);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}

```

OUTPUT:

